1

```
for( i=0, p=0, m=0; i<max; i++) {
    if( i%2==0){
        p++;
    }
    else m++;
}
```

$r_0 \longrightarrow$ adr
$r_1 \longrightarrow$ i
$r_2 \longrightarrow$ p
$r_3 \longrightarrow$ m
$r_4 \longrightarrow$ max

```
init:   MOV  r1, #0
        MOV  r2, #0
        MOV  r3, #0
        ABR  r0, max
        LBR  r4, [r0]

loop:   CMP  r1, r4        i - max
        BGE  end
        AND  r5, r1, #1
        CMP  r5, #0
        ABBEQ r2, r2, #1   -> p++
        ABBNE r3, r3, #1
        ABB  r1, r1, #1
    B loop

end:    ABR  r0, p          ABR r0, m
        STR  r2, [r0]       STR r3, [r0]
```

2.

```
for ( i=0, m=0 ; i < max; i++) {
        m = m+i;
        if( m > 2 * max)   m = 0;
}
```

$r_0 \rightarrow$ adr

$r_1 \rightarrow$ i

$r_2 \rightarrow$ m

$r_3 \rightarrow$ max

```
init:   MOV   r1, #0
        MOV   r2, #0
        ABR   r0, max
        LBR   r3, [r0]
loop:   CMP   r1, r3
        BGE   end
        ADD   r2, r2, r1
        ADD   r4, r3, r3
        CMP   r2, r4
        MOVGT r2, #0
        ADD   r1, r1, #1

        B loop
end:    ABR   r0, m
        STR   r2, [r0]
```

3.

```
for( i=1, s=0 ; i<=n ; i++) {
        if(i %3 == 0) s += i*i ;
}
```

$r_0 \longrightarrow$ adr
$r_1 \longrightarrow$ i
$r_2 \longrightarrow$ s
$r_3 \longrightarrow$ n
$r_4 \longrightarrow$ tmp
$r_5 \longrightarrow$ rest

```
init:   MOV   r1, #1
        MOV   r2, #0
        MOV   r0, n
        MOV   r3, [r0]
        MOV   r5, #1
loop:   CMP   r1, r3
        BGT   end
        CMP   r5, #0
        BNE   skip
        MUL   r4, r1, r1
        ADD   r2, r2, r4        s+= i*i

skip:   ADD   r1, r1, #1
        ADD   r5, r5, #1
        CMP   r5, #3
        MOVEQ r5, #0
        B loop
end:    ADR   r0, s
        STR   r2, [r0]
```